

Data Warehouses and Data Mining Systems

Laboratory 6: Report content

1 Introduction

The sole purpose of the data warehouses is to provide data for analysis. A data warehouse can be thus perceived as a source of valuable data that are cleaned, uniformly formatted and stored in a one place. Data warehouses are modeled using a multi-dimensional approach which means that analyzed events and phenomena (facts) are described with multiple circumstances of occurrence (dimensions). Such a multidimensional approach is also present during analysis – a data warehouse is queried by grouping by dimensions (single attributes or whole hierarchies) and aggregating measures (attributes of a fact table) – e.g. an average price for different cities and years.

Usually data warehouses provide a special view for such multidimensional data – a cube. A cube allows users to view and retrieve data from the data warehouse using a graphical tools (e.g. Excel pivot tables) or a specialized query language (e.g. MDX). A cube formalizes the facts, measures, dimensions, attributes, and hierarchies and allows users to browse data while providing them with high-level operations, such as: pivot, roll-up, drill-down, slice, dice etc. When users browse the cube, several SQL queries are generated by the OLAP server and their results are transformed into the multidimensional form.

OLAP servers and cubes are not always available, especially when a ‘simple’ database management system is concerned (e.g. MySQL). In such scenarios, one have to manually form appropriate SQL queries and transform the result into a readable format.

2 (Re)designing the analyses

Think once again about the analyses that are going to be performed on your data warehouse. Having the data already in the data warehouse enables you to see whether these analyses are feasible or not. Redesign them if necessary. In the report, write down what particular analyses are scheduled for execution.

3 Building the cube (variant A)

If you are using a software that enables you to build and deploy a cube (e.g. Microsoft SQL Server), use it appropriately. Desing a cube (or more than one) and deploy it (them) onto the OLAP server. In the report describe all cubes and their details.

4 Defining queries (variant B)

If your technology stack does not include the possibility of building a cube, you have to manually design and execute SQL queries according to your analyses. Write in the report all queries that you are going to use.

5 Obtaining the results

In order to obtain results, you will be querying the cube or the data warehouse. When cubes are used, you can select two approaches:

1. Use the MDX query language for querying the cube(s) and retrieve the desired data;
2. Browse the cube graphically, e.g. in Excel.

When you are writing your queries directly in the SQL language, you can also use Excel's pivot tables for example but you have to manually prepare data for them.

6 Presentation of results

Whether you use cubes or not, the results have to be properly presented. Generally presenting data as graphs and charts is a good idea. You can even use a dedicated software, like Microsoft BI or Excel for this purpose.